

Problem 3

Part a)

Let the emission angle of the particle be Θ relative to the downward vertical axis (i.e., the line between $(0,0)$ and $(0,p)$). For the particle to hit the x-axis, the emission angle must strictly fall between $-\pi/2$ and $\pi/2$ radians, i.e., $\Theta \in (-\pi/2, \pi/2)$.

The probability space is given by $(\Omega, \mathcal{F}, \mathbb{P})$, where:

- $\Omega = (-\pi/2, \pi/2)$
- $\mathcal{F} = \mathcal{B}(\Omega) = \mathcal{B}((-\pi/2, \pi/2))$ (the Borel σ -algebra on the interval)
- $\mathbb{P}(F) = \frac{\lambda^1(F)}{\lambda^1(\Omega)} = \frac{\lambda^1(F)}{\pi/2 - (-\pi/2)} = \frac{\lambda^1(F)}{\pi} \quad \forall F \in \mathcal{F}$, where λ^1 is the one-dimensional Lebesgue measure.

To calculate $\mathbb{P}(C_{a,b})$, we need to find the probability that the particle hits the x-axis in the spatial interval $[a, b]$. Let X be the random variable representing the impact coordinate. We must find the probability that $X \in [a, b]$ by mapping it to the emission angle Θ . For this, we use the trigonometric ratio $\tan(\Theta) = \frac{\text{opposite}}{\text{adjacent}}$ of the right triangle formed by the source, the origin, and the impact point, which gives the relationship $X = p \tan(\Theta)$.

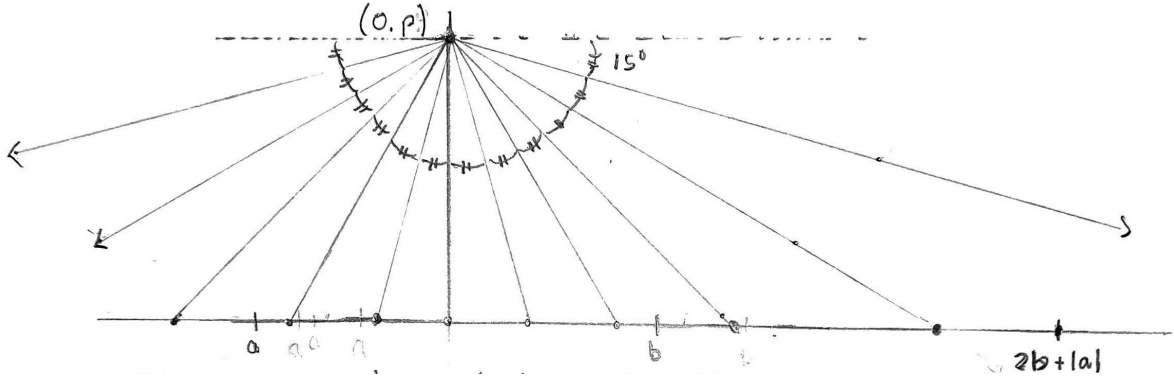


Figure 1: The source emits particles uniformly, but they don't hit the x-axis uniformly. For example, particles are emitted every 15 degrees in the angle range $(-\pi/2, \pi/2)$; 5 of them arrive to the interval $[a, b]$, but only 2 arrive to the interval $[b, 2b + a]$, although their lengths are the same.

Since the relationship between the emission angle Θ and the x-coordinate X is one-to-one and monotonically increasing, the event $\{X \in [a, b]\}$ is strictly equivalent to the event $\left\{ \Theta \in \left[\arctan\left(\frac{a}{p}\right), \arctan\left(\frac{b}{p}\right) \right] \right\}$. Therefore, the probability that the particle hits the x-axis in the interval $[a, b]$ is evaluated using our measure $\mathbb{P}$:

$$\begin{aligned} \mathbb{P}(C_{a,b}) &= \mathbb{P}(X \in [a, b]) = \mathbb{P}\left(\Theta \in \left[\arctan\left(\frac{a}{p}\right), \arctan\left(\frac{b}{p}\right) \right]\right) \\ &= \frac{1}{\pi} \left(\arctan\left(\frac{b}{p}\right) - \arctan\left(\frac{a}{p}\right) \right) \end{aligned}$$

Part b)

We have that:

$$\begin{aligned} F(b) &= \lim_{a \rightarrow -\infty} \mathbb{P}(C_{a,b}) = \lim_{a \rightarrow -\infty} \frac{1}{\pi} \left(\arctan\left(\frac{b}{p}\right) - \arctan\left(\frac{a}{p}\right) \right) \\ &= \frac{1}{\pi} \left(\arctan\left(\frac{b}{p}\right) - \lim_{a \rightarrow -\infty} \arctan\left(\frac{a}{p}\right) \right) = \frac{1}{\pi} \left(\arctan\left(\frac{b}{p}\right) - \left(-\frac{\pi}{2}\right) \right) \\ &= \frac{1}{\pi} \arctan\left(\frac{b}{p}\right) + \frac{1}{\pi} \left(\frac{\pi}{2}\right) = \frac{1}{\pi} \arctan\left(\frac{b}{p}\right) + \frac{1}{2} \end{aligned}$$

We prove three properties for $F(b)$:

1. **Limits:** $\lim_{b \rightarrow -\infty} F(b) = \frac{1}{\pi} \left(-\frac{\pi}{2}\right) + \frac{1}{2} = 0$, and $\lim_{b \rightarrow \infty} F(b) = \frac{1}{\pi} \left(\frac{\pi}{2}\right) + \frac{1}{2} = 1$.
2. **Monotonicity:** Since $p > 0$, the argument $\frac{b}{p}$ is strictly increasing w.r.t. b . Because $\arctan(z)$ is a strictly increasing function on $\mathbb{R}$, their composition is strictly increasing. Multiplying by the positive constant $\frac{1}{\pi}$ and adding $\frac{1}{2}$ preserves this strict monotonicity. Therefore, $F(b)$ is strictly increasing.
3. **Right-continuity:** $F(b)$ is a composition of continuous functions and is therefore continuous on all of $\mathbb{R}$. Hence, it is right-continuous.

Since all three properties hold, $F(b)$ is a cumulative distribution function.

Part c)

We set up the limit:

$$\lim_{x \rightarrow \infty} \frac{1 - F(x)}{\frac{p}{\pi x}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{2} - \frac{1}{\pi} \arctan\left(\frac{x}{p}\right)}{\frac{p}{\pi x}}$$

Differentiate the numerator and denominator with respect to x :

$$\frac{d}{dx} \left(\frac{1}{2} - \frac{1}{\pi} \arctan\left(\frac{x}{p}\right) \right) = -\frac{1}{\pi} \left(\frac{1}{1 + (x/p)^2} \right) \cdot \frac{1}{p} = -\frac{p}{\pi(p^2 + x^2)}$$

$$\frac{d}{dx} \left(\frac{p}{\pi x} \right) = -\frac{p}{\pi x^2}$$

Applying L'Hôpital's rule by differentiating, we evaluate the limit of the quotient of the derivatives:

$$\lim_{x \rightarrow \infty} \frac{-\frac{p}{\pi(p^2 + x^2)}}{-\frac{p}{\pi x^2}} = \lim_{x \rightarrow \infty} \frac{x^2}{p^2 + x^2} = \lim_{x \rightarrow \infty} \frac{1}{\frac{p^2}{x^2} + 1} = 1$$

Because the limit equals 1, the asymptotic equivalence $1 - F(x) \sim \frac{p}{\pi x}$ holds for large x .